

Student Name: _____

Maths Class: _____



James Ruse Agricultural High School

Year 12 Trial HSC Examination 2023

Mathematics Advanced

Reading time: 10 minutes

Working time: 3 hours

General Instructions

- Write using black pen.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- In Questions 11-32, show relevant mathematical reasoning and/or calculations.

Total Marks: 100

Section I

- Attempt questions 1-10.
- Answer on the multiple choice answer sheet provided.
- Allow approximately 15mins for this section.

Sections II, III and IV

- Attempt questions 11-32.
- Answer on the space provided on the booklet. These spaces provide guidance for the expected length of response.
- Allow approximately 2hrs 45mins for these three sections.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Additional writing space is provided at the back of each section. If you use this space, clearly indicate which question you are answering.

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10.

1. Which of the following transformations would produce a graph with the same x -intercepts as $y = f(x)$?

- (A) $y = -f(x)$
- (B) $y = f(-x)$
- (C) $y = f(x + 1)$
- (D) $y = f(x) + 1$

2. Which term of the G.P. is represented by 64 as the last term

$$\left\{ \frac{1}{\sqrt{2}}, 1, \sqrt{2}, \dots, 64 \right\}?$$

- (A) 6
- (B) 9
- (C) 11
- (D) 14

3. What is the equation of the tangent to the curve $y = \sin x$ at the origin?

- (A) $y = -x$
- (B) $y = \cos x$
- (C) $y = \sin x$
- (D) $y = x$

4. $\log_c(a) + \log_a(b) + \log_b(c)$ is equal to

- (A) $\frac{1}{\log_c(a)} + \frac{1}{\log_a(b)} + \frac{1}{\log_b(c)}$
- (B) $\frac{1}{\log_a(c)} + \frac{1}{\log_b(a)} + \frac{1}{\log_c(b)}$
- (C) $-\frac{1}{\log_c(a)} - \frac{1}{\log_a(b)} - \frac{1}{\log_b(c)}$
- (D) $\frac{1}{\log_a(a)} + \frac{1}{\log_b(b)} + \frac{1}{\log_c(c)}$

5. Which of the following can represent the n th term of a GP?

- (i) n^3 (ii) $2^n + 3^n$ (iii) $2^n \times 3^n$ (iv) $3^2 \times 2^n$

- (A) iii only
(B) iii, iv only
(C) i, iii, iv only
(D) i, ii, iii and iv

6. Which of the following is a correct expression for $\frac{d}{dx} \operatorname{cosec}(x)$

- (A) $-\sec(x) \tan(x)$
(B) $-\sec(x) \cot(x)$
(C) $-\operatorname{cosec}(x) \tan(x)$
(D) $-\operatorname{cosec}(x) \cot(x)$

7. Which of the following is a possible solution of $\int \frac{2}{3-4x} dx$

- (A) $\frac{1}{2} \ln|4x - 3| + 2$
(B) $2 \ln|4x - 3| + 2$
(C) $\frac{-1}{2} \ln|4x - 3| + 2$
(D) $-2 \ln|4x - 3| + 2$

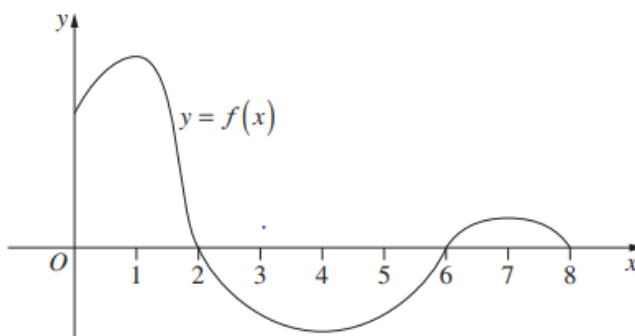
8. Which of the following conditions over the interval $[a, b]$ will result in the trapezoidal rule overestimating the value of the integral $\int_a^b f(x) dx$

- (A) $f(x)$ is increasing
(B) $f(x)$ is decreasing
(C) $f(x)$ is concave up
(D) $f(x)$ is concave down

9. Using the identity $\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$ from the Reference Sheet, or otherwise, identify which expression is equal to $\int \sin^2 3x \, dx$?

- (A) $\frac{1}{2}\left(x - \frac{1}{3}\cos 6x\right) + C$
 (B) $\frac{1}{2}\left(x + \frac{1}{3}\cos 6x\right) + C$
 (C) $\frac{1}{2}\left(x - \frac{1}{6}\sin 6x\right) + C$
 (D) $\frac{1}{2}\left(x + \frac{1}{6}\sin 6x\right) + C$

10. The graph of $y = f(x)$ has been drawn to scale for $0 \leq x \leq 8$.



Which of the following integrals has the greatest value?

- (A) $\int_0^1 f(x) \, dx$
 (B) $\int_0^2 f(x) \, dx$
 (C) $\int_0^7 f(x) \, dx$
 (D) $\int_0^8 f(x) \, dx$

End Section I

Section I

Student Number: _____

Attempt Questions 1 – 10

Allow approximately 15 minutes for this section

Use the multiple-choice answer sheet below to record your answers to Question 1 – 10.

Select the alternative: A, B, C or D that best answers the question.

Colour in the response oval completely.

Sample:

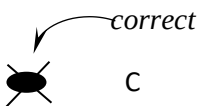
$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, draw a cross through the incorrect answer and colour in the new answer

ie A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word “correct” and draw an arrow as follows:

A ☒ B ☒ C ☐ D ☐


Year 12 Trial 2023

Mathematics Advanced Trial HSC Examination

Multiple Choice Answer Sheet

Completely colour in the response oval representing the most correct answer.

1	A	☐	B	☐	C	☐	D	☐
2	A	☐	B	☐	C	☐	D	☐
3	A	☐	B	☐	C	☐	D	☐
4	A	☐	B	☐	C	☐	D	☐
5	A	☐	B	☐	C	☐	D	☐
6	A	☐	B	☐	C	☐	D	☐
7	A	☐	B	☐	C	☐	D	☐
8	A	☐	B	☐	C	☐	D	☐
9	A	☐	B	☐	C	☐	D	☐
10	A	☐	B	☐	C	☐	D	☐

Mark: /10

Section II (30 marks)

Answer in the space provided.

Student Number: _____

11. Justify whether $f(x) = (x - 1)^2$ is an even function.

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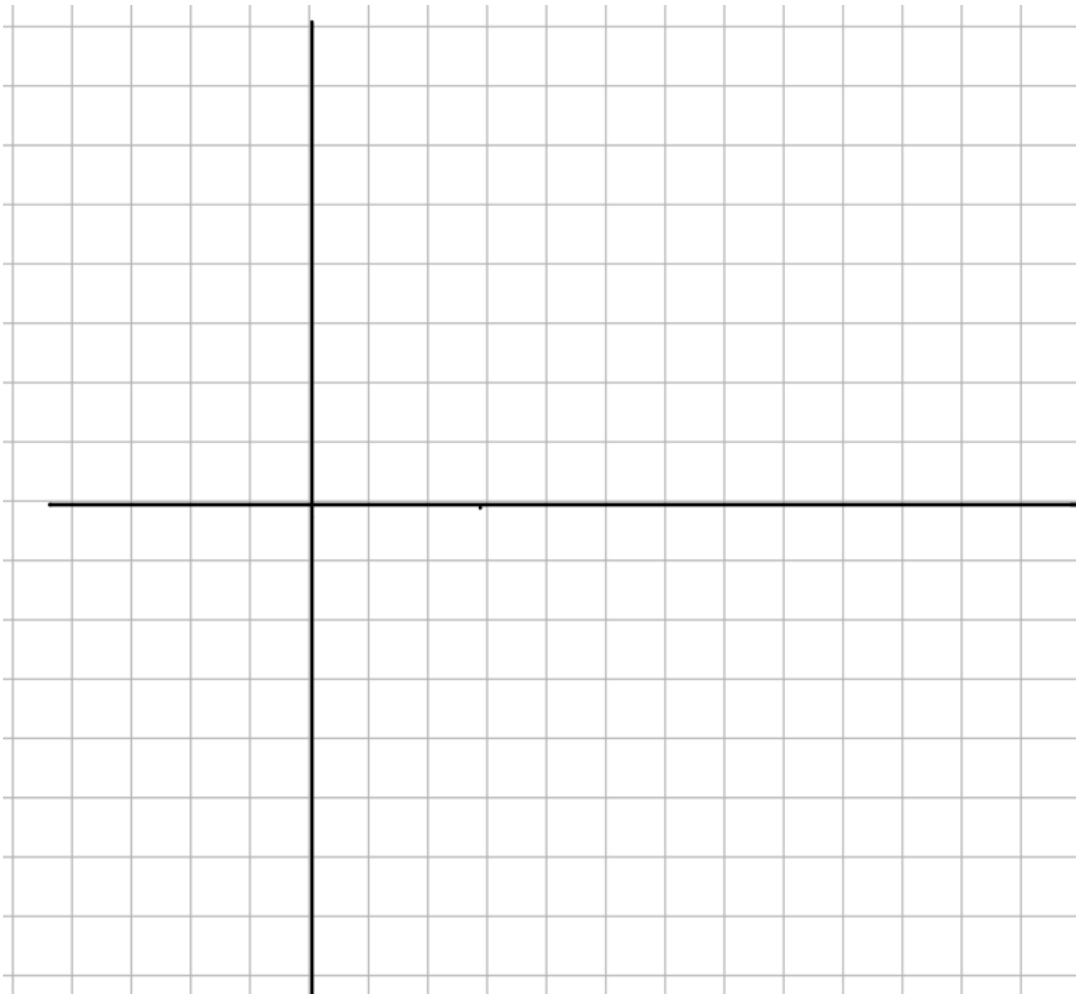
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12. Sketch $y = 3\log_{10}(x - 2)$ showing all important features.

3



13. Solve $|4x - 8| = 3|x + 2|$

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14. The point $P(-1, 3)$ lies on the curve with equation $y = f(x)$.
State the coordinates of the image of point P (called P') after the following transformations have taken place:

3

i. $y = f(x + 3) + 3$

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ii. $y = 2f(x) + 3$

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iii. $y = f(2x + 3)$

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15. Show whether $g(x) = \frac{x-1}{x+1}$ is monotonic increasing or decreasing.

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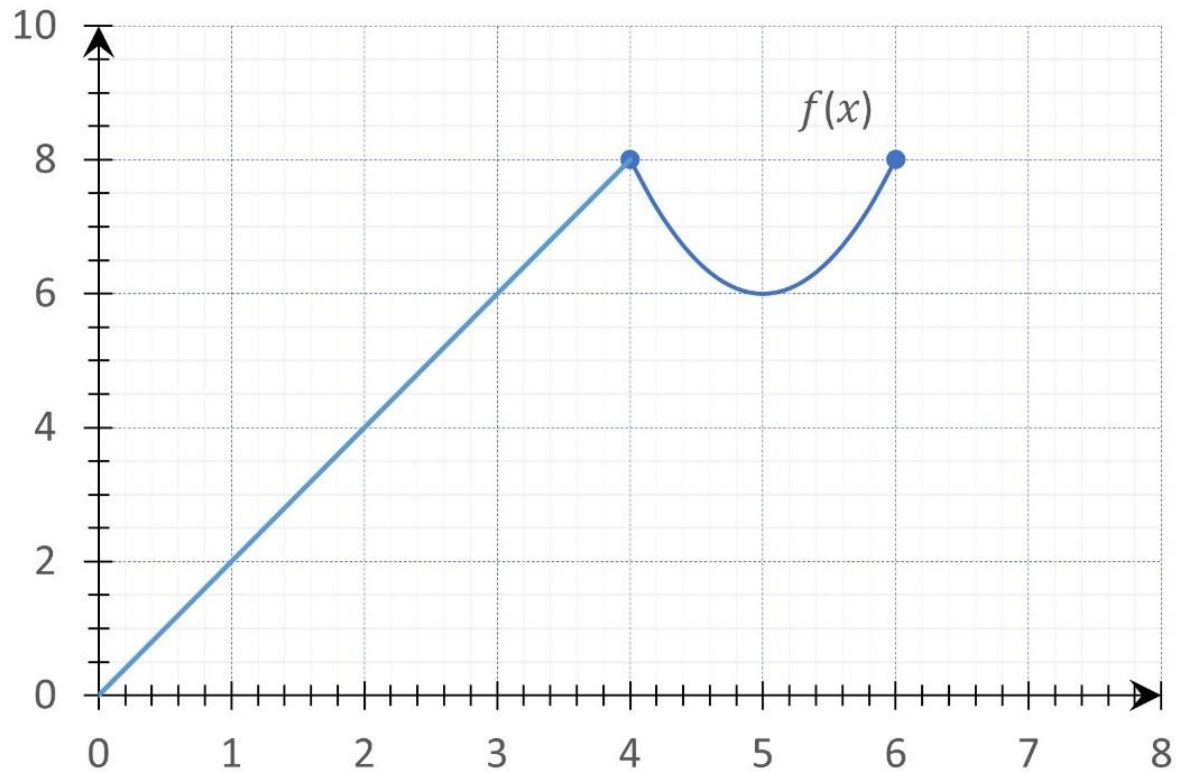
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16. Given the graph of $f(x)$ as shown in the diagram below, sketch on the same axes $g(x)$ and $h(x)$.
Label your graphs clearly.

3

i. $g(x) = f(2x)$.

ii. $h(x) = 0.5f(x)$



17. For the random variable X , it is known that $E(X) = 6$ and $\text{Var}(X) = 2$.
Write down $E(3X - 1)$ for the new distribution $3X - 1$

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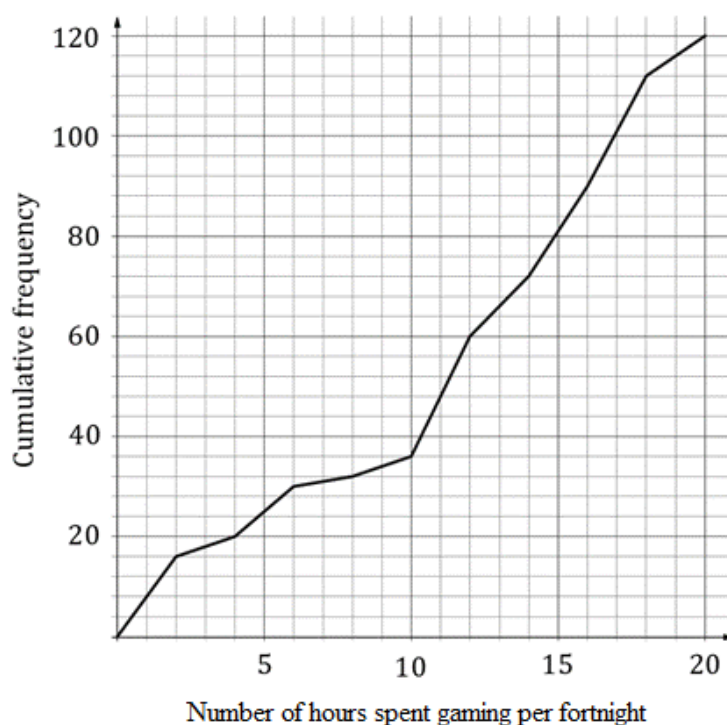
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18. The following cumulative frequency curve shows the number of hours spent gaming per fortnight by 120 high school students.



- (a) Find the Median number of hours spent gaming.

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- (b) Find the Interquartile range.

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- (c) Calculate the percentage of students that spent less than 17 hours gaming per fortnight.

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- (d) Calculate the maximum number of hours that can be spent gaming per fortnight and not be considered an outlier.

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19. Let $f(x) = \left(\frac{1}{2}x - 2\right)^3 + 2$

(a) Show the equation of the tangent to $f(x)$ at the point where $x = 6$ is $y = \frac{3}{2}x - 6$

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(b) On the Graph paper below, graph the functions $f(x)$ and the tangent of $f(x)$ at $x = 6$, showing all important features

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- 12 -

Section III (30 marks)

Student Number: _____

Answer in the space provided.

20. Find all solutions for $\cos x + \sin^2 x = \frac{5}{4}$, $-\pi \leq x \leq \pi$

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21. Find the derivatives of the following :

a. $y = \frac{x}{x^2-5}$

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b. $y = x^2\sqrt{4x-1}$

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22. Find $\int 7x^2 \sin(2x^3 - 5) dx$

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23. A function has equation $y = e^{2x}$. Find the equation of the normal to the curve at the point where $x = 1$.

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24. A continuous random variable has the probability density function given by:

$$f(x) = \begin{cases} ax^3, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a

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(b) Sketch the graph of $f(x)$

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(c) Calculate

$$P(X < \frac{2}{3} \mid X > \frac{1}{3})$$

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25. The population L in thousands of mountain lions, who are predators, and the population D in thousands of deer, who are the prey, in a particular continuously monitored region can be modeled by the following two periodic functions.

$$L = 9 + 6 \sin \frac{\pi}{4}(t - 1)$$

$$D = 18 + 9 \cos \frac{\pi}{4}t$$

Where t is the time in years after the Tracking with Technology started, L and D are the populations (in thousands) of lions and deer respectively.

- (a) How much time passes between successive maximum lions' populations (the period)? 2

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- (b) Determine the initial populations of lions and deer respectively. 2

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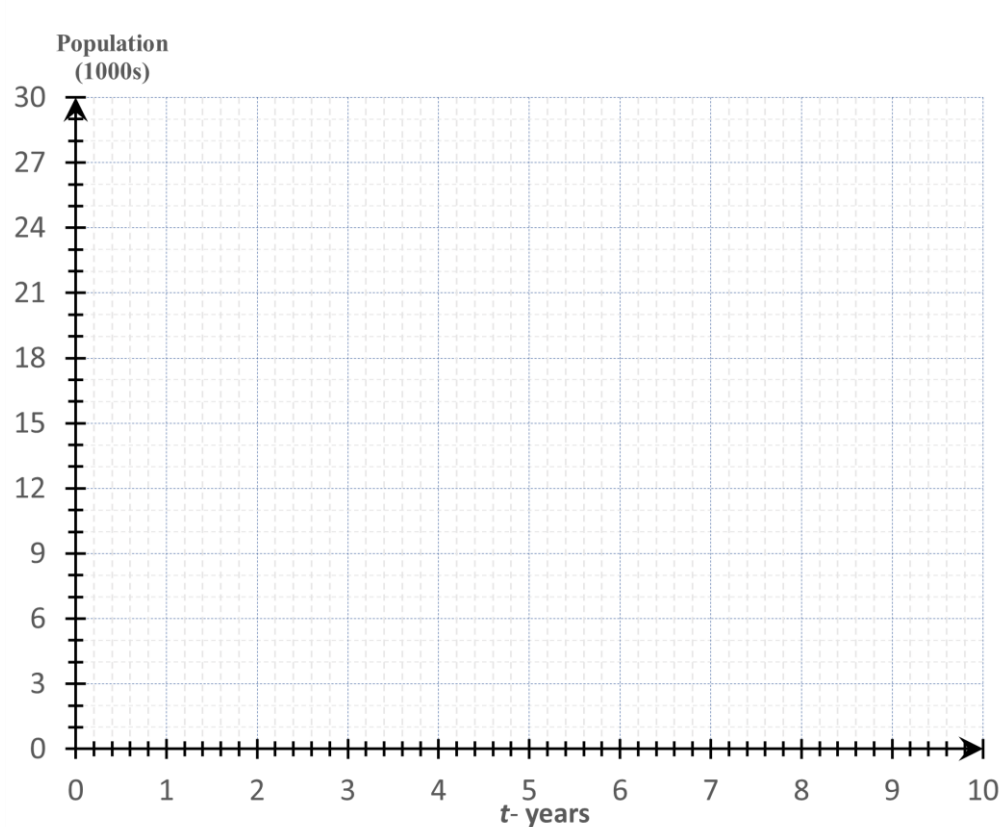
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- (c) Sketch the graphs of the two functions (the Lion and Deer functions) on the same axes in the domain $0 \leq t \leq 10$ 2



- (d) Over which time period, $0 \leq t \leq 10$, was the deer population increasing while the lion population decreased? 1

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26. Solve $\ln x - \frac{6}{\ln x} = 5$

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27. If $x^2 = (2a - x)(2b - x)$ show that $\frac{1}{a}$, $\frac{1}{x}$, $\frac{1}{b}$ are in arithmetic sequence.

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End Section III

Section IV (30 marks)

Student Number: _____

Answer in the space provided.

28. Katie has deliberately designed a biased six sided die, with the following probability distribution for X , the number on the uppermost face when the die is rolled.

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1 - \theta}{6}$	$\frac{1 - \theta}{6}$	$\frac{1 - \theta}{6}$	$\frac{1 + \theta}{6}$	$\frac{1 + \theta}{6}$	$\frac{1 + \theta}{6}$

- (a) What values of θ are feasible in order for $P(X)$ to be a probability function?

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- (b) Find $P(0 \leq X \leq 4)$

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- (c) Find the probability of rolling an even number?

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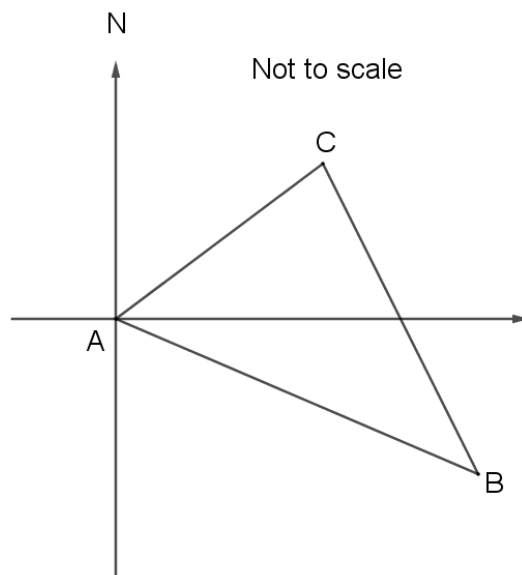
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29. A student on a bike left point A and walked 1.5km on a bearing of 120°T to a point B. He then turned and walked 10km on a bearing of 330°T to a point C.



- (a) Label all features on the diagram above. 2

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- (b) Calculate the distance from point C to point A. (give your answer to 2 decimal places) 2

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- (c) What is the bearing of A from C ? (Give your answer to the nearest degree) 2

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30. A particle is moving in a straight line. Initially, it is travelling to the left at 1cm/min.
Its acceleration is given by:

$$a = \pi \cos (\pi t) + \pi \sin (\pi t)$$

for $0 \leq t \leq 2$ where time and displacement are measure in minutes and cm respectively.

- (a) Find when the particle first changes its direction.

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- (b) Find the total distance travelled in the first half a minute.

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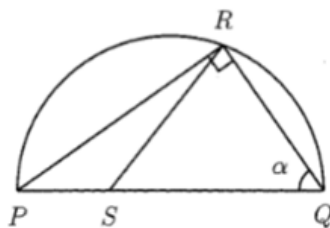
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31. $\triangle PQR$ is a right-angled triangle inscribed in a semi-circle. R is a variable point on the circumference. The point S lies on PQ such that $SQ = kQR$ where k is a positive constant. If $PQ = d$ cm and $\angle PQR = \alpha$ radians:



- (a) Show that the area of $\triangle SQR$ is $A = \frac{1}{2}kd^2\cos^2\alpha\sin\alpha$.

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- (b) Show that $\frac{dA}{d\alpha} = \frac{1}{2}kd^2(3\cos^3\alpha - 2\cos\alpha)$.

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(c) Find the greatest possible area of ΔSQR in terms of k and d .

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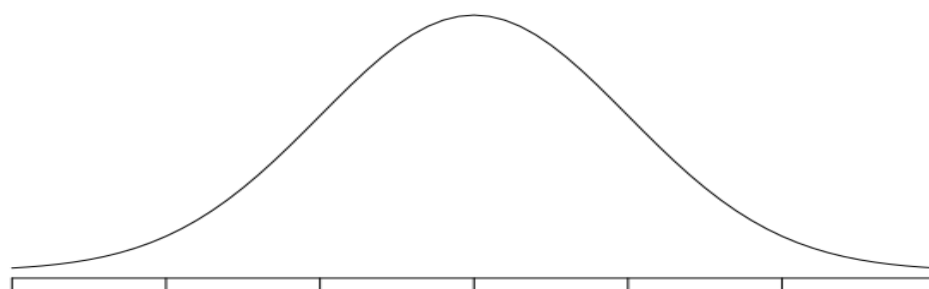
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32. An airline company operates regular flights between cities A and B. The flight time X is normally distributed with a mean of 80 minutes and standard deviation σ minutes. Also, 2.5% of the flights take longer than 96 minutes to arrive at the destination.

(a) Label the normal distribution curve below with the given information.

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(b) Find the value of σ .

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(c) What percentage of flights take longer than 90 minutes?

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(d) In order to attract customers, the management of the airline company decides to refund the fares if a flight takes longer than $(80 + t)$ minutes. The company wants to keep 99.5% of the fares. Calculate the value of t .

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Year 12 Trial HSC Examination 2023

Mathematics Advanced

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10 marks

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Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10.

1. Which of the following transformations would produce a graph with the same x -intercepts as $y = f(x)$?

(A) $y = -f(x)$ *no change*

(B) $y = f(-x)$ *$x \rightarrow -x$*

(C) $y = f(x + 1)$ *$x \rightarrow x+1$*

(D) $y = f(x) + 1$ *x moves*

2. Which term of the G.P. is represented by ^{64 as} the last term of the set

$$\left\{ \frac{1}{\sqrt{2}}, 1, \sqrt{2}, \dots, 64 \right\}?$$

(A) 6

(B) 9

(C) 11

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3. What is the equation of the tangent to the curve $y = \sin x$ at the origin?

(A) $y = -x$

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(B) $2 \ln|4x - 3| + 2$
(C) $-\frac{1}{2} \ln|4x - 3| + 2$
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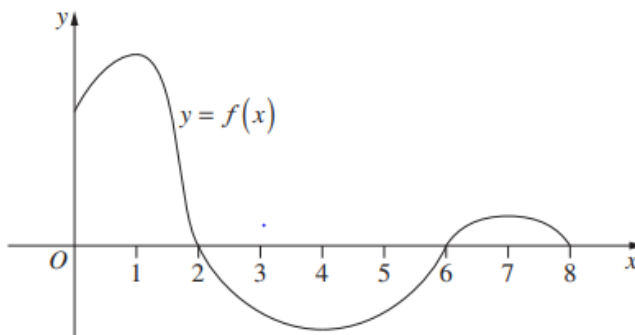
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(B) $\frac{1}{2}\left(x + \frac{1}{3}\cos 6x\right) + C$
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10. The graph of $y = f(x)$ has been drawn to scale for $0 \leq x \leq 8$.



Which of the following integrals has the greatest value?

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End Section I

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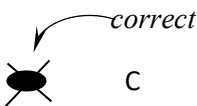
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A ☐ B ☒ C ☐ D ☐

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A ☒ B ☒ C ☐ D ☐


Year 12 Trial 2023

Mathematics Advanced Trial HSC Examination

Multiple Choice Answer Sheet

Completely colour in the response oval representing the most correct answer.

1	A	☒	B	☐	C	☐	D	☐
2	A	☐	B	☐	C	☐	D	☒
3	A	☐	B	☐	C	☐	D	☒
4	A	☐	B	☒	C	☐	D	☐
5	A	☐	B	☒	C	☐	D	☐
6	A	☐	B	☐	C	☐	D	☒
7	A	☐	B	☐	C	☒	D	☐
8	A	☐	B	☐	C	☒	D	☐
9	A	☐	B	☐	C	☒	D	☐
10	A	☐	B	☒	C	☐	D	☐

Mark: /10

Section II (30 marks)

Answer in the space provided.

Student Number: _____

11. Justify whether $f(x) = (x - 1)^2$ is an even function.

2

$$\text{even if } f(x) = f(-x)$$

$$f(x) = (x - 1)^2 = x^2 - 2x + 1$$

$$f(-x) = (-x - 1)^2 = x^2 + 2x + 1$$

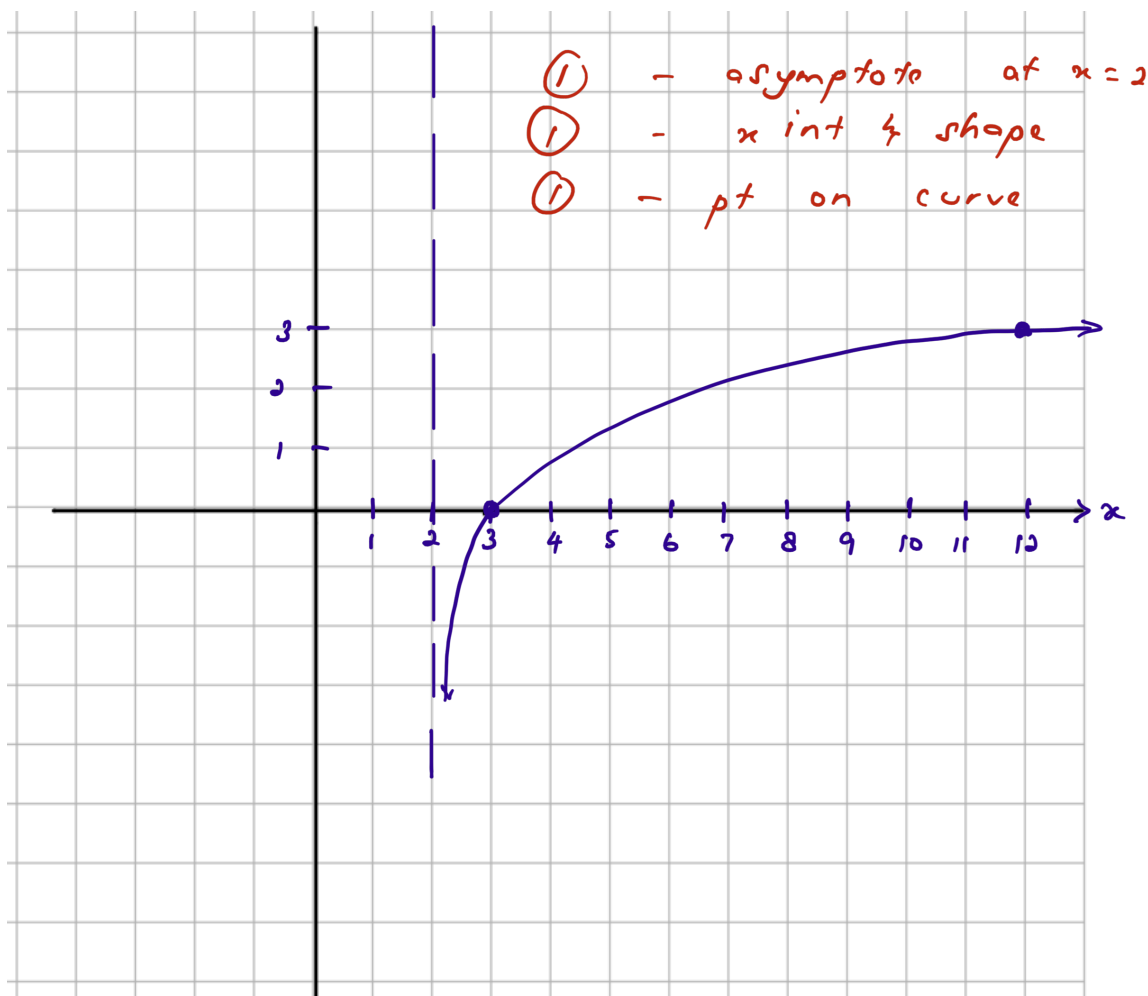
$$f(x) \neq f(-x)$$

$$\therefore f(x) \text{ is not an even func}$$

} show — ①
 state — ①

12. Sketch $y = 3\log_{10}(x - 2)$ showing all important features.

3



Ex + T3
Done

13. Solve $|4x - 8| = 3|x + 2|$

2

Method 1

$$4x - 8 = 3(x + 2)$$

$$4x - 8 = 3x + 6$$

$$x = 14$$

$$4x - 8 = -3(x + 2)$$

$$4x - 8 = -3x - 6$$

$$7x = 2$$

$$x = \frac{2}{7}$$

Method 2 Square both sides

$$16x^2 - 64x + 64 = 9(x^2 + 4x + 4)$$

$$16x^2 - 64x + 64 = 9x^2 + 36x + 36$$

$$7x^2 - 100x + 28 = 0$$

$$x = \frac{-(-100) \pm \sqrt{(-100)^2 - 4(7)(28)}}{2(7)}$$

$$2(7)$$

$$x = \frac{2}{7} \text{ or } 14$$

14. The point $P(-1, 3)$ lies on the curve with equation $y = f(x)$.

3

State the coordinates of the image of point P (called P') after the following transformations have taken place:

i. $y = f(x + 3) + 3$

$$x \text{ left } 3 \therefore P' = (-1 - 3, 3 + 3)$$

$$y \text{ up } 3 = (-4, 6)$$

ii. $y = 2f(x) + 3$

$$y \Rightarrow x + 3 \quad P' = (-1, 2 \times 3 + 3)$$

$$= (-1, 9)$$

iii. $y = f(2x + 3)$

$$x \Rightarrow \text{compressed ie } x \frac{1}{2} \quad P' = \left(-\frac{1}{2} - \frac{3}{2}, 3\right)$$

$$\text{then shift left by } \frac{3}{2} = (-2, 3)$$

$$f\left(2\left(x + \frac{3}{2}\right)\right)$$

15. Show whether $g(x) = \frac{x-1}{x+1}$ is monotonic increasing or decreasing.

2

M1

$$u = x - 1, \quad u' = 1$$

$$v = x + 1, \quad v' = 1$$

$$g'(x) = \frac{(x+1) \cdot 1 - (x-1) \cdot 1}{(x+1)^2}$$

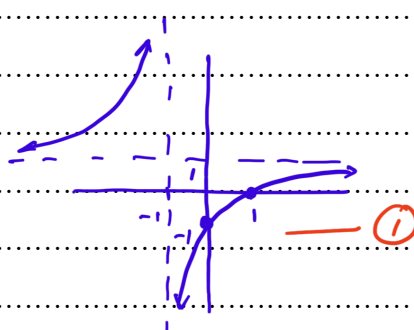
show

$$\frac{2}{(x+1)^2}$$

$$(x+1)^2 > 0 \text{ then } f'(x) > 0$$

state thus $f(x)$ is monotonic increasing

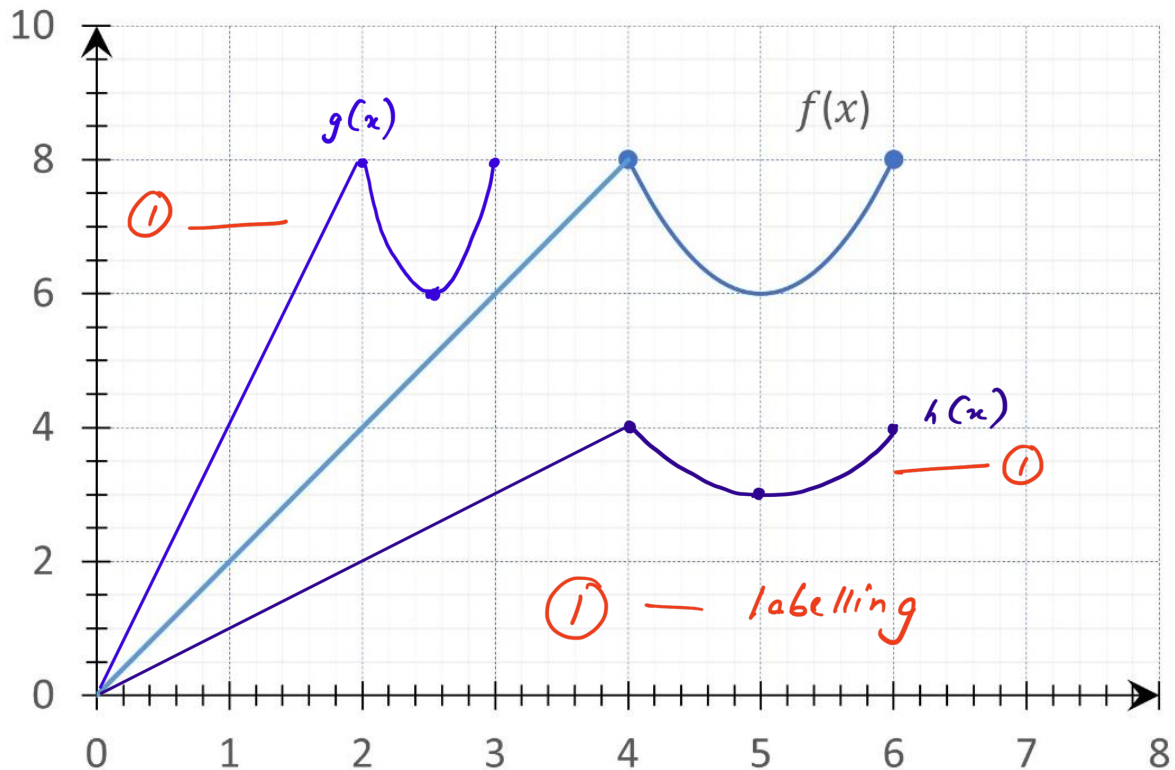
M2



then state

16. Given the graph of $f(x)$ as shown in the diagram below, sketch on the same axes $g(x)$ and $h(x)$. Label your graphs clearly. 3

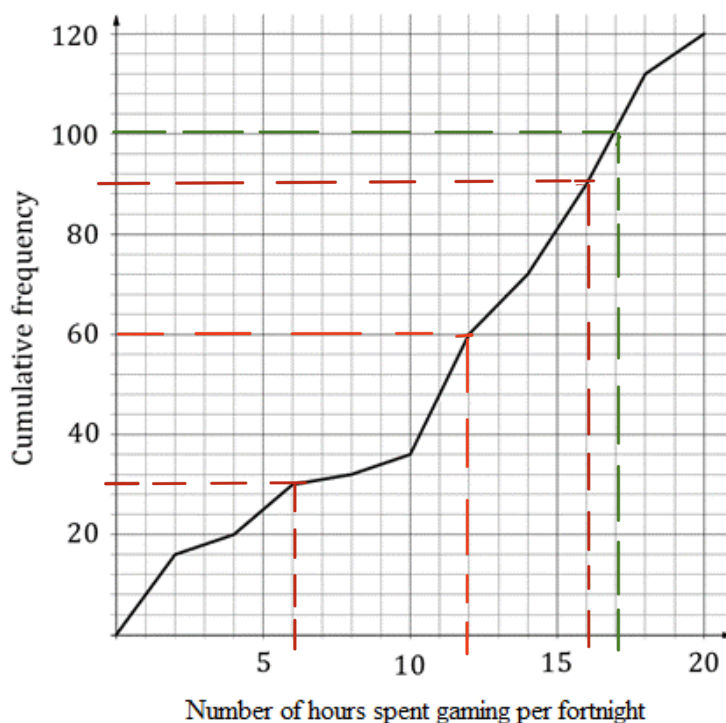
- i. $g(x) = f(2x)$. *horizontal dilation $x \rightarrow x \frac{1}{2}$*
 ii. $h(x) = 0.5f(x)$ *vertical dilation $y \rightarrow y \frac{1}{2}$*



17. For the random variable X , it is known that $E(X) = 6$ and $\text{Var}(X) = 2$.
 Write down $E(3X - 1)$ for the new distribution $3X - 1$ 1

$$\begin{aligned}
 E(X) &= 6 \\
 E(3X - 1) &= 3(E(X)) + (-1) \\
 &= 3(6) - 1 \\
 &= 17 \quad \text{--- (1)}
 \end{aligned}$$

18. The following cumulative frequency curve shows the number of hours spent gaming per fortnight by 120 high school students.



- (a) Find the Median number of hours spent gaming.

1

12 hours

①

- (b) Find the Interquartile range.

2

16 - 6

①

= 10

①

- (c) Calculate the percentage of students that spent less than 17 hours gaming per fortnight.

1

$$\frac{100}{120} \times 100\% = 83.3\%$$

①

- (d) Calculate the maximum number of hours that can be spent gaming per fortnight and not be considered an outlier.

2

Max no of hrs = ?

$Q_3 + 1.5 \text{ IQR}$

(from reference sheet)

16 + 15

= 31 ie 31 is an outlier

①

Not to be an outlier is < 31

①

19. Let $f(x) = \left(\frac{1}{2}x - 2\right)^3 + 2$

(a) Show the equation of the tangent to $f(x)$ at the point where $x = 6$ is $y = \frac{3}{5}x - 6$

2

$$\begin{aligned}
 f'(x) &= 3\left(\frac{1}{2}x - 2\right)^2 \cdot \frac{1}{2} \\
 &= \frac{3}{2}\left(\frac{1}{2}x - 2\right)^2 \\
 f'(6) &= \frac{3}{2}\left(\frac{1}{2}(6) - 2\right)^2 \\
 &= \frac{3}{5} \quad \text{--- (1)}
 \end{aligned}$$

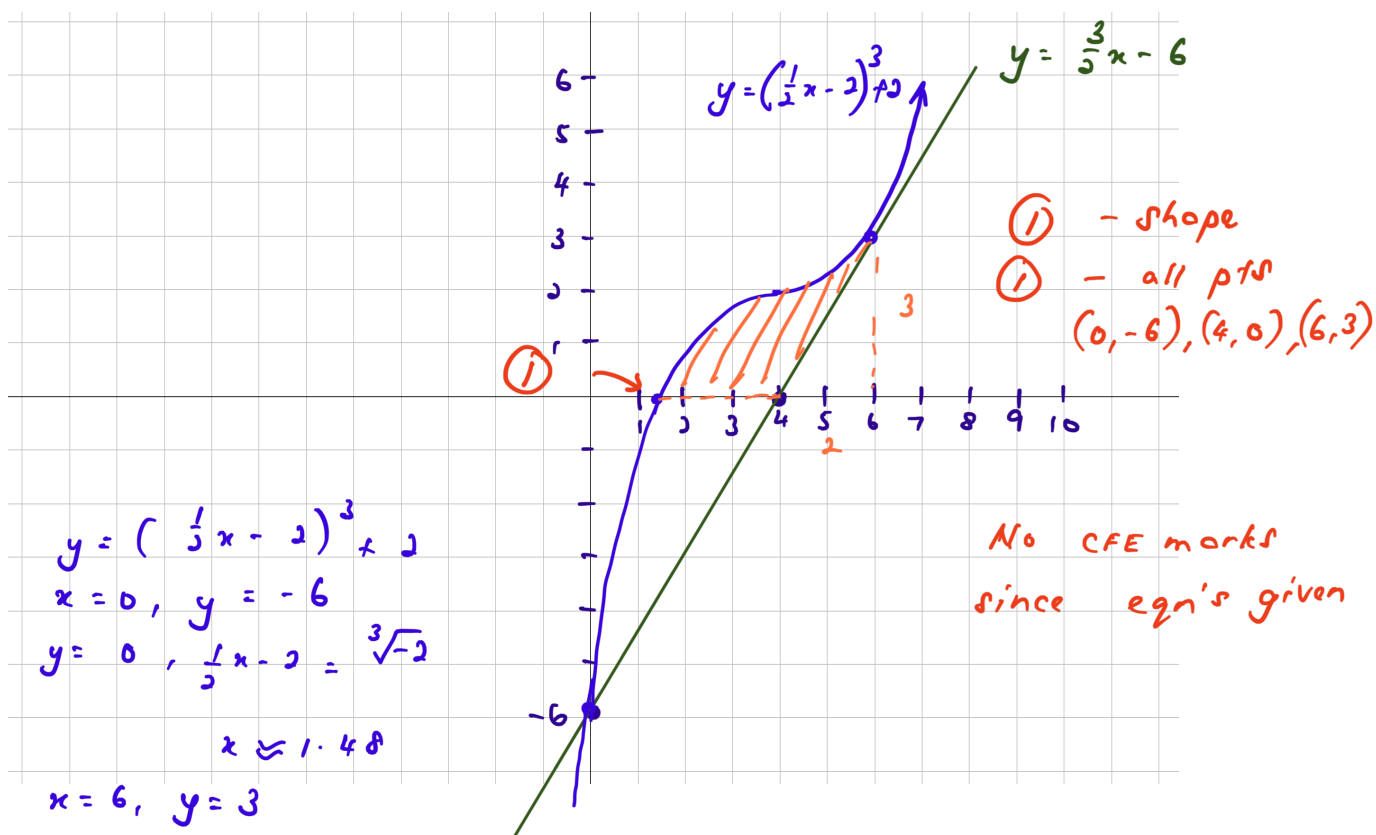
when $x = 6$, $y = 3$

$$\begin{aligned}
 \therefore \text{Eqn of tangent} &\Rightarrow y - 3 = \frac{3}{5}(x - 6) \\
 &y = \frac{3}{5}x - 9 + 3 \\
 &y = \frac{3}{5}x - 6 \quad (\text{shown})
 \end{aligned}$$

Poorly done
Yr d content
given a pt & gradient

(b) On the Graph paper below, graph the functions $f(x)$ and the tangent of $f(x)$ at $x = 6$, showing all important features

3



- (c) Calculate the area above the x-axis and enclosed between $f(x)$ and the tangent, where $0 < x < 6$.

3

$$= \int_{1.48}^6 \left(\frac{1}{5}x - 2 \right)^3 \pi \, dx - A \text{ of } \Delta \quad \text{--- (1)}$$

$$= \int_{1.48}^6 \frac{\left(\frac{1}{5}x - 2 \right)^4}{4 \cdot \frac{1}{5}} \pi \, dx - \frac{1}{2} \times 3 \times 2$$

--- (1) *integrating correctly*

$$= 5.279 \dots$$

$$= 5.3 \text{ (1 d.p.)}$$

} --- (1)

End Section II

Section III (30 marks)
Answer in the space provided.

Student Number: _____

20. Find all solutions for $\cos x + \sin^2 x = \frac{5}{4}$, $-\pi \leq x \leq \pi$

3

$$\begin{aligned} \cos x + (1 - \cos^2 x) &= \frac{5}{4} && 1 \text{ mark} \\ \cos^2 x - \cos x + \frac{1}{4} &= 0 \\ 4\cos^2 x - 4\cos x + 1 &= 0 \\ (2\cos x - 1)^2 &= 0 && 1 \text{ mark} \\ 2\cos x - 1 &= 0 \\ 2\cos x &= 1 \\ \cos x &= \frac{1}{2} \\ \therefore x &= \frac{\pi}{3} \text{ or } x = -\frac{\pi}{3} && 1 \text{ mark} \end{aligned}$$

21. Find the derivatives of the following:

a. $y = \frac{x}{x^2 - 5}$

2

$$\begin{aligned} \frac{dy}{dx} &= \frac{(x^2 - 5) - x(2x)}{(x^2 - 5)^2} && 1 \text{ mark for using quotient rule formula correctly. } u = x \quad u' = 1 \\ &= \frac{x^2 - 5 - 2x^2}{(x^2 - 5)^2} \\ &= \frac{-x^2 - 5}{(x^2 - 5)^2} && 1 \text{ mark for correct answer. } v = x^2 - 5 \quad v' = 2x \end{aligned}$$

b. $y = x^2 \sqrt{4x - 1}$

2

$$\begin{aligned} \frac{dy}{dx} &= (4x - 1)^{\frac{1}{2}} \cdot 2x + x^2 \left(\frac{2}{(4x - 1)^{\frac{1}{2}}} \right) && 1 \text{ mark for substituting into Product Rule formula. } u = x^2 \quad u' = 2x \\ &= \frac{2x(4x - 1) + 2x^2}{(4x - 1)^{\frac{1}{2}}} && v = (4x - 1)^{\frac{1}{2}} \\ &= \frac{8x^2 - 2x + 2x^2}{(4x - 1)^{\frac{1}{2}}} && v' = \frac{1}{2}(4x - 1)^{-\frac{1}{2}} \cdot 4 \\ &= \frac{10x^2 - 2x}{(4x - 1)^{\frac{1}{2}}} && = 2(4x - 1)^{-\frac{1}{2}} \\ &= \frac{10x^2 - 2x}{(4x - 1)^{\frac{1}{2}}} && = \frac{2}{(4x - 1)^{\frac{1}{2}}} \\ &= \frac{10x^2 - 2x}{(4x - 1)^{\frac{1}{2}}} && 1 \text{ mark for answer} \end{aligned}$$

22. Find $\int 7x^2 \sin(2x^3 - 5) dx$

$$= \frac{7}{6} \int 6x^2 \sin(2x^3 - 5) dx$$

1 mark for $\cos(2x^3 - 5)$

$$= -\frac{7}{6} \cos(2x^3 - 5) + c$$

1 mark for $-\frac{7}{6}$

1 mark for +c if
most of rest is correct.

23. A function has equation $y = e^{2x}$. Find the equation of the normal to the curve at the point where $x = 1$.

$$y = e^{2x}$$

$$y' = 2e^{2x}$$

$$\text{When } x=1 \quad y = e^2$$

$$y' = 2e^2$$

Since the gradient of tangent is $2e^2$ then

gradient of normal is $-\frac{1}{2e^2}$

1 mark for gradient
of normal

Equation of Normal is: $y - e^2 = -\frac{1}{2e^2}(x - 1)$

$$2e^2 y - 2e^2 \cdot e^2 = -x + 1$$

$$x + 2e^2 y - 2e^4 - 1 = 0$$

$$\underline{\text{OR}} \quad y = \frac{-1}{2e^2}(x-1) + e^2$$

$$= \frac{-x+1}{2e^2} + \frac{2e^4}{2e^2}$$

$$= \frac{-x + 2e^4 + 1}{2e^2}$$

1 mark for equation
in either format.

24. A continuous random variable has the probability density function given by:

$$f(x) = \begin{cases} ax^3, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a

2

$$\int_0^1 ax^3 dx = 1$$

$$\left[\frac{ax^4}{4} \right]_0^1 = 1$$

$$\frac{a}{4} - 0 = 1$$

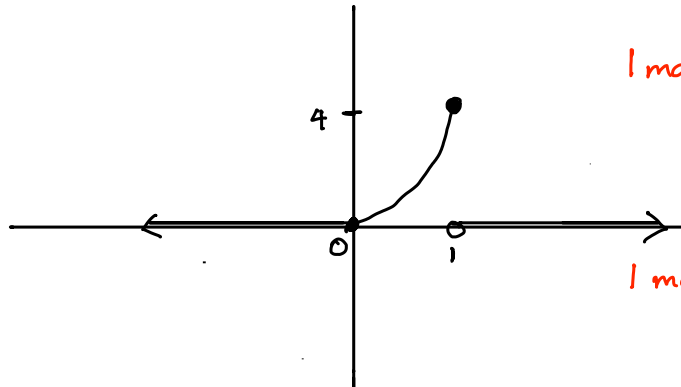
$$a = 4$$

1 mark for correct integration and = 1

1 mark for answer.

(b) Sketch the graph of $f(x)$

3



1 mark for ax^3 between 0 and 1 on x-axis and 0 to 4 on y-axis.

1 mark for open/closed circles in correct place

1 mark for $x > 1$ and $x < 0$

(c) Calculate

$$P(X < \frac{2}{3} | X > \frac{1}{3})$$

3

$$\frac{P(A \cap B)}{P(B)} = \frac{P(X < \frac{2}{3} \cap X > \frac{1}{3})}{P(X > \frac{1}{3})}$$

1 mark for understanding & using conditional probability

$$= \frac{\int_{\frac{1}{3}}^{\frac{2}{3}} 4x^3 dx}{\int_{\frac{1}{3}}^1 4x^3 dx}$$

OR $F(x) = \int_0^x 4x^3 dx$
 $= x^4$

$$P(X > \frac{1}{3}) = 1 - F(\frac{1}{3})^4$$

$$= \frac{80}{81}$$

$$= \frac{\int_{\frac{1}{3}}^1 4x^3 dx}{1 - \frac{1}{81}}$$

$$= \frac{(\frac{2}{3})^4 - (\frac{1}{3})^4}{1 - \frac{1}{81}}$$

1 mark.

$$= \frac{\frac{16}{81} - \frac{1}{81}}{1 - \frac{1}{81}}$$

$$P(X < \frac{2}{3}) = F(\frac{2}{3})$$

$$= (\frac{2}{3})^4 = \frac{16}{81}$$

- 15 -

$$= \frac{\frac{15}{81}}{\frac{80}{81}}$$

$$= \frac{15}{80}$$

1 mark for

or $\frac{3}{16}$ correct answer.

$$P(X < \frac{1}{3}) = (\frac{1}{3})^4 = \frac{1}{81}$$

25. The population L in thousands of mountain lions, who are predators, and the population D in thousands of deer, who are the prey, in a particular continuously monitored region can be modeled by the following two periodic functions.

$$L = 9 + 6 \sin \frac{\pi}{4}(t - 1)$$

$$D = 18 + 9 \cos \frac{\pi}{4}t$$

Where t is the time in years after the Tracking with Technology started, L and D are the populations (in thousands) of lions and deer respectively.

- (a) How much time passes between successive maximum lions' populations (the period)? 2

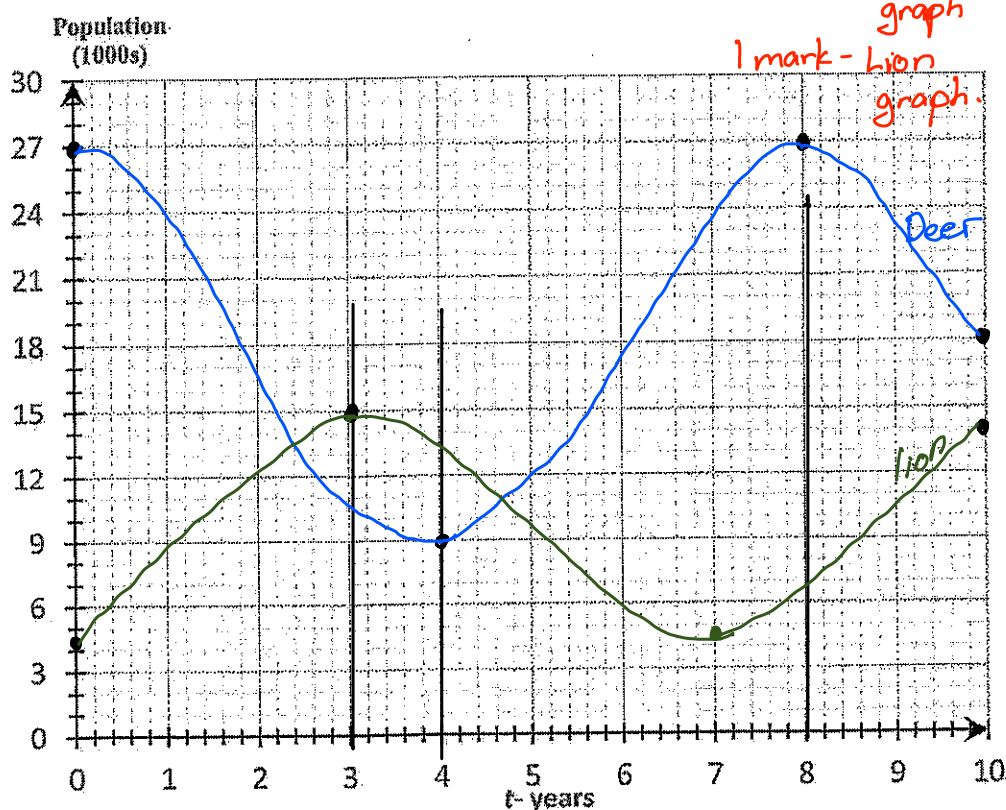
$$\begin{aligned} L &= 9 + 6 \sin \frac{\pi}{4}(t - 1) \\ \text{Period} &= \frac{2\pi}{h} \\ &= \frac{2\pi}{\frac{\pi}{4}} \quad \text{1 mark} \\ &= 8 \text{ years} \quad \text{1 mark.} \end{aligned}$$

- (b) Determine the initial populations of lions and deer respectively. 2

$$\begin{aligned} \text{When } t &= 0 \\ L &= 9 + 6 \sin \frac{\pi}{4}(-1) & D &= 18 + 9 \cos \left(\frac{\pi}{4} \times 0 \right) \\ &= 9 + 6 \times -\frac{1}{\sqrt{2}} & &= 18 + 9 \\ &= 9 - 3\sqrt{2} & &= 27 \text{ thousand} \\ &\approx 4.757 \text{ thousand} & &= 27000 \quad \text{1 mark.} \\ &= 4.757 \text{ lions} \quad \text{1 mark} \\ \therefore \text{Initially there were } &4.757 \text{ lions} \\ &\text{and } 27000 \text{ deer.} \end{aligned}$$

- (c) Sketch the graphs of the two functions (the Lion and Deer functions) on the same axes in the domain $0 \leq t \leq 10$

2



- (d) Over which time period, $0 \leq t \leq 10$, was the deer population increasing while the lion population decreased?

1

$4 < t < 7$ from graph.

1 mark

must have a graph.

26. Solve $\ln x - \frac{6}{\ln x} = 5$

2

$$(\ln x)^2 - 6 = 5 \ln x$$

$$(\ln x)^2 - 5 \ln x - 6 = 0$$

$$\text{Let } u = \ln x$$

$$u^2 - 5u - 6 = 0$$

$$x^{-6}$$

1 mark

$$(u-6)(u+1) = 0$$

$$u = 6 \text{ or } u = -1$$

$$\therefore \ln x = 6 \text{ or } \ln x = -1$$

$$x = e^6 \text{ or } x = \frac{1}{e}$$

1 mark.

$T_1 \quad T_2 \quad T_3$

27. If $x^2 = (2a - x)(2b - x)$ show that $\frac{1}{a}, \frac{1}{x}, \frac{1}{b}$ are in arithmetic sequence.

3

$$x^2 = 4ab - 2ax - 2bx + x^2$$

$$4ab - 2ax - 2bx = 0$$

$$4ab - 2x(a+b) = 0$$

$$2x = \frac{4ab}{a+b}$$

$$x = \frac{2ab}{a+b}$$

$$\frac{1}{x} = \frac{a+b}{2ab}$$

1 mark

$$\therefore T_2 - T_1 = \frac{a+b}{2ab} - \frac{1}{a}$$

$$= \frac{a+b-2b}{2ab}$$

$$= \frac{a-b}{2ab}$$

1 mark

$$\text{and } T_3 - T_2 = \frac{1}{b} - \frac{a+b}{2ab}$$

$$= \frac{2a - (a+b)}{2ab}$$

$$= \frac{a-b}{2ab}$$

$$\text{Since } T_2 - T_1 = T_3 - T_2 = \frac{a-b}{2ab}$$

1 mark for clear reason.

then $\frac{1}{a}, \frac{1}{x}, \frac{1}{b}$ are in arithmetic sequence.

End Section III

Q 29 / 6

Q 30, 31 / 13

Q 28, 32 / 11

Section IV (30 marks)

Answer in the space provided.

Student Number: _____

28. Katie has deliberately designed a biased six sided die, with the following probability distribution for X , the number on the uppermost face when the die is rolled.

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1-\theta}{6}$	$\frac{1-\theta}{6}$	$\frac{1-\theta}{6}$	$\frac{1+\theta}{6}$	$\frac{1+\theta}{6}$	$\frac{1+\theta}{6}$

- (a) What values of θ are feasible in order for $P(X)$ to be a probability function?

2

$$3\left(\frac{1-\theta}{6}\right) + 3\left(\frac{1+\theta}{6}\right) = 1 \quad (1)$$

$$\frac{6 - 3\theta + 3\theta}{6} = 1$$

$$\frac{6}{6} = 1 \quad \therefore \text{works for all } \theta$$

$$\text{But } \frac{1-\theta}{6} \geq 0 \quad \text{and} \quad \frac{1+\theta}{6} \geq 0$$

$$\theta \leq 1 \quad \text{and} \quad \theta \geq -1$$

$$\therefore -1 \leq \theta \leq 1 \quad (1)$$

- (b) Find $P(0 \leq X \leq 4)$

1

$$3\left(\frac{1-\theta}{6}\right) + \frac{1+\theta}{6}$$

$$= \frac{4-2\theta}{6} = \frac{2-\theta}{3} \quad (1)$$

- (c) Find the probability of rolling an even number?

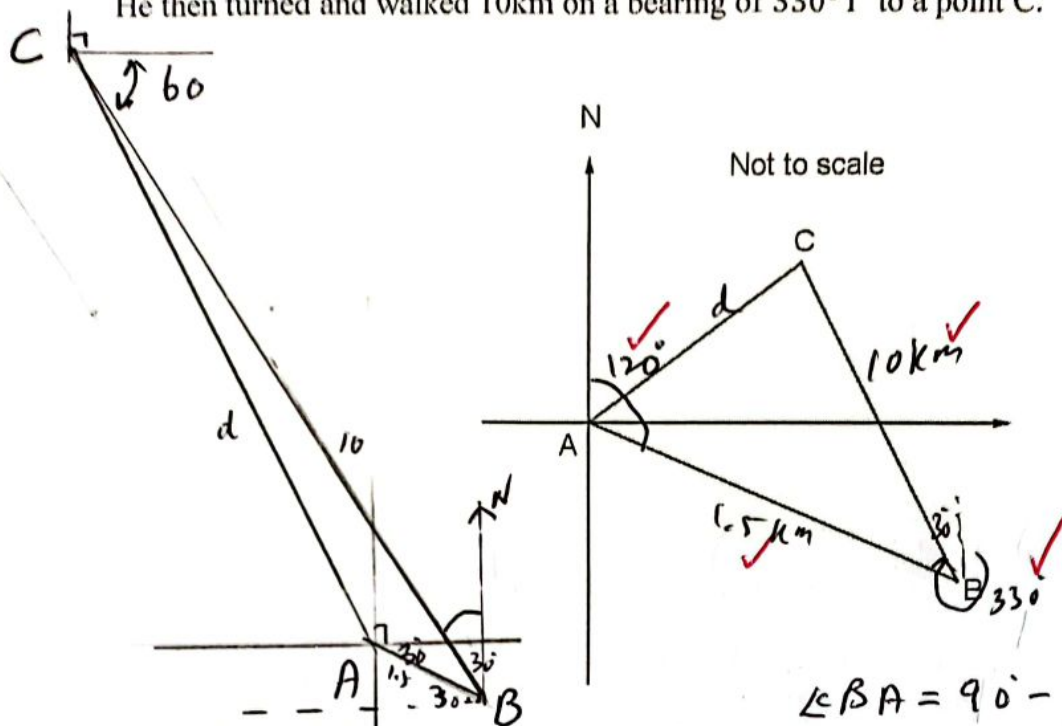
1

$$P(2) + P(4) + P(6)$$

$$= \frac{1-\theta}{6} + \frac{2(1+\theta)}{6}$$

$$= \frac{3+\theta}{6} = \frac{1}{2} + \frac{\theta}{6} \quad (1)$$

29. A student on a bike left point A and walked 1.5km on a bearing of $120^{\circ}T$ to a point B. He then turned and walked 10km on a bearing of $330^{\circ}T$ to a point C.



any 2 ✓ 1 m

$$\angle B A = 90^{\circ} - 30^{\circ} - 30^{\circ} = 30^{\circ}$$

- (a) Label all features on the diagram above.

2

- (b) Calculate the distance from point C to point A. (give your answer to 2 decimal places)

2

$$d^2 = 10^2 + 1.5^2 - 2(10)(1.5)\cos 30^{\circ}$$

$$d^2 = 102.25 - 30 \times \frac{\sqrt{3}}{2}$$

$$= 76.2692$$

$$d = 8.7332 \text{ (d > 0)}$$

$$d = 8.73 \text{ (2 d.p.)}$$

km

- (c) What is the bearing of A from C? (Give your answer to the nearest degree)

2

$$1.5^2 = (8.7332)^2 + 10^2 - 2(8.7332)(10)\cos \angle A C B$$

$$\cos \angle A C B = \frac{176.2692 - 2.25}{174.664}$$

$$= 0.9963$$

$$\angle A C B = 4.9247^{\circ} = 4^{\circ}55' \approx 5^{\circ} \text{ (nearest degree)}$$

$$\text{Bearing of A from C} = 90^{\circ} + 60^{\circ} + 5^{\circ} = 155^{\circ} T$$

- 20 -

Closest degree

30. A particle is moving in a straight line. Initially, it is travelling to the left at 1 cm/min. Its acceleration is given by:

$$a = \pi \cos(\pi t) + \pi \sin(\pi t)$$

for $0 \leq t \leq 2$ where time and displacement are measured in minutes and cm respectively.

- (a) Find when the particle first changes its direction.

2

$$v = \sin(\pi t) - \cos(\pi t) + c$$

$$t=0 \quad v=-1 \quad 0-1+c=-1$$

$$c=0$$

$$\therefore v = \sin(\pi t) - \cos(\pi t) \quad (1)$$

change direction when $v=0$

$$\therefore \sin(\pi t) = \cos(\pi t)$$

$$\tan(\pi t) = 1$$

$$\pi t = \frac{\pi}{4}, \frac{5\pi}{4} \text{ etc.} \quad (1)$$

1st change direction when $t = \frac{1}{4} \text{ min}$

- (b) Find the total distance travelled in the first half a minute.

3

$$x = \left| \int_0^{\frac{1}{4}} (\sin \pi t - \cos \pi t) dt \right| + \int_{\frac{1}{4}}^{\frac{1}{2}} (\sin \pi t - \cos \pi t) dt \quad (1m)$$

$$x = \left| \left(-\frac{\cos \pi t}{\pi} - \frac{\sin \pi t}{\pi} \right) \right|_0^{\frac{1}{4}} + \left(-\frac{\cos \pi t}{\pi} - \frac{\sin \pi t}{\pi} \right)_{\frac{1}{4}}^{\frac{1}{2}} \quad (1m)$$

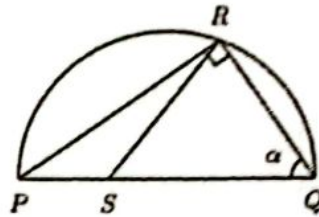
$$= \left| \frac{1}{\pi} \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right) \right| + \frac{1}{\pi} \left(\cos \frac{\pi}{2} + \sin \frac{\pi}{2} \right) - \left(\frac{\cos \frac{\pi}{4}}{\pi} + \frac{\sin \frac{\pi}{4}}{\pi} \right)$$

$$= \left| \frac{1}{\pi} \left(-\frac{\sqrt{2}}{2} + 1 \right) - \frac{\pi}{\pi} + \frac{2}{\sqrt{2}} \right|$$

$$= \frac{1}{\pi} \left[(\sqrt{2}-1) - 1 + \sqrt{2} \right] = 0.2637 (4dp)$$

$$= \frac{2\sqrt{2}-2}{\pi} \quad (1m)$$

31. ΔPQR is a right-angled triangle inscribed in a semi-circle. R is a variable point on the circumference. The point S lies on PQ such that $SQ = kQR$ where k is a positive constant. If $PQ = d$ cm and $\angle PQR = \alpha$ radians:



- (a) Show that the area of ΔSQR is $A = \frac{1}{2}kd^2\cos^2\alpha\sin\alpha$.

3

$$\begin{aligned}
 \text{Area } \Delta SQR &= \frac{1}{2}SQ \cdot QR \sin \alpha \\
 &= \frac{1}{2}(kQR)(QR) \sin \alpha \quad \text{1 m} \\
 &= \frac{1}{2}k(QR)^2 \sin \alpha \quad \text{1 m} \quad (QR = d \cos \alpha) \\
 &= \frac{1}{2}k(d \cos \alpha)^2 \sin \alpha \quad \text{1 m} \quad (SQ = kd \cos \alpha) \\
 &= \frac{1}{2}kd^2 \cos^2 \alpha \sin \alpha
 \end{aligned}$$

- (b) Show that $\frac{dA}{d\alpha} = \frac{1}{2}kd^2(3\cos^3\alpha - 2\cos\alpha)$.

2

$$\begin{aligned}
 \frac{dA}{d\alpha} &= \frac{1}{2}kd^2 \frac{d}{d\alpha} (\cos^2 \alpha \sin \alpha) \\
 &= \frac{1}{2}kd^2 \left[\cos^2 \alpha \cos \alpha + (\sin \alpha) 2 \cos \alpha (\sin \alpha) \right] \quad \text{1 m} \\
 &= \frac{1}{2}kd^2 (\cos^3 \alpha - 2 \cos \alpha (\sin^2 \alpha)) \\
 &= \frac{1}{2}kd^2 (\cos^3 \alpha - 2 \cos \alpha (1 - \cos^2 \alpha)) \quad \text{1 m} \\
 &= \frac{1}{2}kd^2 (\cos^3 \alpha - 2 \cos \alpha + 2 \cos^3 \alpha) \\
 &= \frac{1}{2}kd^2 (3 \cos^3 \alpha - 2 \cos \alpha)
 \end{aligned}$$

(c) Find the greatest possible area of ΔSQR in terms of k and d .

3

Greatest possible area when $\frac{1}{2}kd^2(3\cos^2\alpha - 2\cos\alpha) = 0$

Since $k > 0, d > 0$

$$\cos\alpha(3\cos\alpha - 2) = 0$$

$$\therefore \cos\alpha = 0 \text{ or } \cos\alpha = \frac{2}{3}$$

$$\cos\alpha = 0 \text{ or } \cos\alpha = \pm\sqrt{\frac{2}{3}}$$

Since $0 < \alpha < \frac{\pi}{2} \therefore \cos\alpha = \sqrt{\frac{2}{3}}$ only

1 m

1 m for $0 < \alpha < \frac{\pi}{2}$

$\therefore \cos\alpha \neq 0$
 $\cos\alpha \neq \sqrt{\frac{2}{3}}$

when

$$\cos\alpha = \sqrt{\frac{2}{3}}$$

$$\alpha = 0.61547$$

$$\begin{aligned} \text{Area} &= \frac{1}{2}kd^2\cos\alpha\sin\alpha = \frac{1}{2}kd^2 \cdot \frac{2}{3} \sqrt{\frac{1}{3}} \\ &= \frac{kd^2}{3\sqrt{3}} \end{aligned}$$

check max/min

α	0.5	0.61547	1
A	0.7724 $\times \frac{1}{2}kd^2$	0	-0.6074 $\times \frac{1}{2}kd^2$

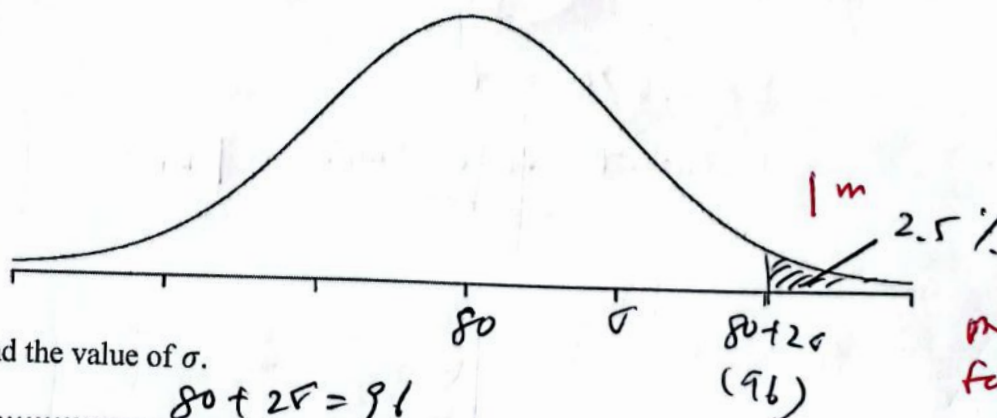
$\therefore$ max A at $\cos\alpha = \sqrt{\frac{2}{3}}$

1 m for max A & check

32. An airline company operates regular flights between cities A and B. The flight time X is normally distributed with a mean of 80 minutes and standard deviation σ minutes. Also, 2.5% of the flights take longer than 96 minutes to arrive at the destination.

(a) Label the normal distribution curve below with the given information.

1



(b) Find the value of σ .

$$80 + 2\sigma = 96$$

$$\sigma = 8$$

many forget 2.5%

(1 m)

(c) What percentage of flights take longer than 90 minutes?

2

$$z = \frac{90 - 80}{8} = 1.25 \quad | \text{ m}$$

10.56% from z table
| m

same answer

as 0.1056 get 1 m

(d) In order to attract customers, the management of the airline company decides to refund the fares if a flight takes longer than $(80 + t)$ minutes. The company wants to keep 99.5% of the fares.

3

$$0.57 = 0.005$$

$$z = 2.575 \quad | \text{ m}$$

$$z = \frac{x - \bar{x}}{s}$$

$$2.575 = \frac{80 + t - 80}{8} \quad | \text{ m}$$

$$8 \times 2.575 = t$$

$$t = 20.6 \text{ min} \quad | \text{ m}$$